

Crypto MCAL用户手册

Crypto MCAL用户手册

1. 密钥使用注意
2. 同步方式使用注意
3. 修改BusClock
4. 真随机数
 - 4.1 NVR配置
5. RSA签名验签
 - 5.1 RSA私钥
 - 5.1.1 上位机脚本
 - 5.1.2 CRYPTO_KE_FORMAT_BIN_IDENT_PRIVATEKEY_PKCS8
 - 5.1.2.1 配置
 - 5.1.2.2 使用

1. 密钥使用注意

密钥空间里的密钥尽量在产线通过烧录工具烧好，就不再改动。产线上是直接写Flash来烧录密钥数据。

对密钥空间里的key，慎用 `Crypto_KeySetValid`，根据AUTOSAR规范，每次调用这个API，都需要更新密钥空间里的密钥内容，这对密钥空间的寿命有损伤。

由于密钥空间是DFlash，所以设置密钥为无效，只是program一些标志位，需要用户手动触发去整理密钥空间。但每次整理密钥空间，都是erase，然后重新program。erase会损耗Flash的寿命。

如果调用 `Crypto_KeySetValid` 设置NVM密钥空间里的密钥数据，先调用 `Crypto_KeySetInvalid` 把所有需要 `Crypto_KeySetValid`设置密钥数据的密钥都统一先设置为无效。

如果更新密钥，先废弃revoke密钥，对于FC7300F8MDT，FC7300F4MDx和FC7240F2MDS，需要先整理密钥空间，然后再重新导入相同key id的密钥。对于FC7300F8MDQ，可以直接更新，内部逻辑也是先废弃revoke，然后追加program。CryptoDriver底层封装了这套逻辑，通常建议用户使用的时候，如果调用 `Crypto_KeySetValid` 设置NVM密钥空间里的密钥数据，先调用 `Crypto_KeySetInvalid` 把所有需要设置密钥数据的密钥都统一先设置为无效。这个过程中要小心突然断电等异常情况的发生。

`Crypto_KeySetInvalid` 在密钥空间里revoke所有需要更新密钥的密钥，然后 `Crypto_KeySetValid` 分别设置特定key id的密钥，只会在设置第一个的时候整理密钥空间，后面都不会整理密钥空间了，这将对DFlash的损伤降到最低。


```

//HSM Software Reset
SCM->SUBSYS_PCC|=SCM_SUBSYS_PCC_SWRST(1);
//Clear HSM Software Reset
SCM->SUBSYS_PCC&=~SCM_SUBSYS_PCC_SWRST(1);
}

static void Bsp_HSM_Reset(void)
{
    //HSM Software Reset
    SCM->SUBSYS_PCC|=SCM_SUBSYS_PCC_SWRST(1);
    //Clear HSM Software Reset
    SCM->SUBSYS_PCC&=~SCM_SUBSYS_PCC_SWRST(1);
}

```

以上代码，涉及到 SCM 的相关宏定义，具体宏定义实现可以查看 SDK 中。详细代码如下，

```

static void Bsp_HSM_Init(void)
{
    //Enable HSM clock, SCM_HSM_PCC register
    *((volatile uint32*)0x40072110) |= ((uint32)0x1u << 23u);
    //HSM Software Reset, SCM_HSM_PCC register
    *((volatile uint32*)0x40072110) |= ((uint32)0x1u << 16u);
    //Clear HSM Software Reset, SCM_HSM_PCC register
    *((volatile uint32*)0x40072110) &= (~(uint32)0x1u << 16u));
}

static void Bsp_HSM_Reset(void)
{
    //HSM Software Reset, SCM_HSM_PCC register
    *((volatile uint32*)0x40072110) |= ((uint32)0x1u << 16u);
    //Clear HSM Software Reset, SCM_HSM_PCC register
    *((volatile uint32*)0x40072110) &= (~(uint32)0x1u << 16u));
}

```

4. 真随机数

4.1 NVR配置

对真随机数配置可以依据 FC7300 手册《FC7300 Reference Manual.pdf》描述，修改 NVR 中 OSCA，OSCFREQ 寄存器的值，确认 SEC 中对应的值 因为 SEC 寄存器组的部分寄存器中是从 NVR 中加载而来，因此需要先修改 NVR，reset 板子，再查看 SEC 寄存器是否正常 load. 操作如下，在 FC7300 中 NVR0 的区域是 0x04400000-0x04407FFF,其中对应的 0x04400080 地址的 bit8-bit10 为 OSCFREQ 寄存器，bit11 为 OSCA 寄存器。NVR 中 0x04400080 地址映射对应手册中的第 21 章节 21.5.13 SEC_BRC1 寄存器，地址是 0x4001406C。手册描述如下，

21.5.13 SEC_BRC1 (BootROM Configuration Register1)

Value loaded from NVR sector. Refer to the NVR section for detailed mapping information.

Offset : 6Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R								SW_C								
								FG_I								
								SP_E								
								N								
W																
RST	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DBG_MB_BKD_EN				OSCA	OSCFREQ			CANBR		UARTBR		FW_VALID			
W																
RST	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

10-8 OSCFREQ	RO	OSC Frequency 000b - 8MHz 001b - 12MHz 010b - 16MHz 011b - 24MHz 100b - 25MHz 101b - 32MHz 110b - 48MHz 111b - 40MHz
11 OSCA	RO	OSC Available 0b - OSC is available 1b - OSC is not available

用户根据自己的板子上的晶振设置对应的值。在此之前， bit11(OSCA)必须为 0。

具体操作为：

使用 jlink.exe 打开 Jlink 终端 -> 输入指令 Connect-> 输入指令一个问号 ' ? ' -> 选择 device FC7300F8MDTXXXXT1B. 如下图。

```

Type "connect" to establish a target connection, '?' for help
J-Link>connect
Please specify device / core. <Default>: FC7300F8MDTXXXXT1B
Type '?' for selection dialog
Device>
Please specify target interface:
  J) JTAG (Default)
  S) SWD
  T) cJTAG
TIF>
Device position in JTAG chain (IRPre,DRPre) <Default>: -1,-1 => Auto-detect
JTAGConf>
Specify target interface speed [kHz]. <Default>: 4000 kHz
Speed>
Device "FC7300F8MDTXXXXT1B" selected.

Connecting to target via JTAG
ConfigTargetSettings() start
ConfigTargetSettings() end
InitTarget() start

```

使用指令查看对应地址 `mem32 04400080 1`

修改 NVR 的值((oscfreq=40Mhz osca=0)) `w4 04400080 FFFF7FFF`

输入 q 指令，则退出终端

板子 reset

```

Cortex-M7 identified.
J-Link>w4 04400080 ffff7fff
Writing FFFF7FFF -> 04400080
J-Link>

```

板子重启以后，查看 NVR 值 `mem32 04400080 1` 和查看 SEC 寄存器 `mem32 4001406c 1`，比较两个值是否相同。

```

J-Link>mem32 4001406c 1
4001406C = FFFF7FFF

```

```

J-Link>mem32 4001406c 1
4001406C = FFFF7FFF
J-Link>mem32 04400080 1
04400080 = FFFF7FFF
J-Link>

```

注意：NVR mass erase 后默认值是 40MHz，需要和实际晶振保持一致。

5. RSA签名验签

5.1 RSA私钥

```
# 生成私钥
openssl genrsa -out id_rsa_private 2048
# 生成私钥(把RSA私钥转换成PKCS8格式)
openssl pkcs8 -topk8 -inform PEM -in id_rsa_private -outform pem -nocrypt -out
id_rsa_private_pkcs
# 生成公钥 pkcs8格式
openssl rsa -in id_rsa_private -pubout -out id_rsa_public.pub
# 生成公钥 pkcs1格式
openssl rsa -pubin -in id_rsa_public.pub -RSAPublicKey_out -out pkcs1_public_key.pem
```

5.1.1 上位机脚本

python安装包

```
pip install cryptography
```

下面就是加密后解密的脚本，可以修改公私钥证书，获取特定测试数据。注意打印里的Key size是否和EB配置是否一致。

```
from cryptography.hazmat.backends import default_backend
from cryptography.hazmat.primitives.asymmetric import padding
from cryptography.hazmat.primitives import serialization
from cryptography.hazmat.primitives import hashes

# 假设这是你的RSA公钥，通常从证书中获取
public_key_pem = """-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAuwbxAmHUBHySm6GSomn0
BcQX3lJLP2d3137Lwk97ZbcGAufZpCGkZXD8fh6ErdJDDJMiwxFSwTrFs7DPHFFwt
YCHSHgf+xJ+NC/ovD1PtXb1Q1b1NTlXWBC1YXaOyAH011iZn0myoFiw7rBbZUMMw
5oeEEDgQGTHfc778i3LQ4TZLH7N+veKWQ2N2s+pLX5r6XEqh3vTMqzBypnuQG0ei
92qQ7YYhsB9Mqk1pmS2QLr71PFjz2vOv4e/VsG6AzkwSn9zf4jgvOrOYYf62013d
BG0g+1VJMQG1JFdpCYts3IGQ45nzyPCDxxlkkVB3z/uC56QCR9SDPSF+KyyFV7aJ
JwIDAQAB
-----END PUBLIC KEY-----"""

# 加载公钥
public_key = serialization.load_pem_public_key(public_key_pem.encode(),
backend=default_backend())

# 需要加密的数据
data = b"secret data"
```

```

# 使用OAEP进行加密
#encrypted = public_key.encrypt(data,
padding.OAEP(mgf=padding.MGF1(algorithm=hashes.SHA256()), algorithm=hashes.SHA256(),
label=None))

# 使用 PKCS1v1.5 进行加密
print("key size:" + str(public_key.key_size))
encrypted = public_key.encrypt(data, padding.PKCS1v15())

# 现在 `encrypted` 包含加密的数据，可以发送给持有私钥的人
print("plain text:" + data.hex())
print("encrypted text:" + encrypted.hex())

#####
#####
# 假设这是你的私钥，通常以-----BEGIN PRIVATE KEY-----开头
private_key_pem = """-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBAKggSkAgEAAoIBAQC7BvECyDRsfJKb
oZKgyfQXfBfeUks/Z3fxfstat3tltwYC59mkIaRlCPx+Host0kMkyJbEVLBOswzs
M8cUXC1hwdIEb/7En40L+i8PU+ldvVDVUu10vdYFzVhdo7IAC7XWJmfSbkGwJbus
Ft1SYzDmh4QQOBAZMd9zvvyLctDhNksfs3694pZDY3az6ktfmvpcSqHe9MyrMHKm
e5AbR6L3apDthiFIH0yqTwmZLZAuvvU8WNna86/h79wwboDOTBKf3N/iOC86s5hh
/rbTXd0EY6D7VukxAbukv2kji1LcgZDjmdnI9wPFewSRUHfP+4LnPAJH1IM9IX4r
LIVXtoknAgMBAAECggEBAIBRSCZCxEI833wiIwTSC5n3nTKiaeY5FCY++mL5f26
Y0Dw9HhWL/VX7wrs4P2T1gbl+bJpAfax+1aBNH6+u4/+cYJobQQ7pyb9H7NC4pzy
A7UssEdJpM/16E8H0yZ0Qp2ubaTH4pqkdJZmmh3Dltp20YIHh8NBqjv7bnzvNRdC
GJpvvbhPhF+jTnbwrKn98gFN2d0EdlQZA5QFs1l8Sw3hwUHHg5JpcTTkohYbn91J
sm33HB6U23Defk52CkG008jqb7NwbtfIO020480AoZShC5Hj4pazHFyZ4kkql+lh
kdkCr1zR6ZAp7Dziym1ak84g+/590tltjx3bl8TSFgECgYEA9HFNA95zLyJwb0pz
65FnTAAVCOCT4t6Kad76yx5AGvkjuEpuZOKaazFRGIq8H0ha6a5UjIdavknyUNWJ
mC9T6P1TdI6zxgFGGRPKDjHrqPTZN3M9/7bKwWRCmj669eI+o6xiaALDLnspAGgx
UXisyefmKG6/jIQA4M2QmKQl6oECgYEAw96xFlG8IYAv3HM4mfUjiym9CuF+heS2
svMK3SOHVivCQtdvaBFj9SftgyUQMSB4vzRFeCjGBfh60T7kCarrvDi0UAE5mJhi
BCUCFMroxBEWS3kozyX8gh1dkr1couzhKbVnVq5vXPqzxZ+YYJJkyZrd+p/waVZL
j+Awp2m9D6cCgYEA1xiGIYhsqQw5LONCEAc582mbGN2tpJQYsr7Ixed/d3zGdtQI
QmQYZ7Zu1Ji4IWPhJAIQSjYmpKobQ8AzvKpg3YOVHVq5HmJgMY71XPpwma0joyUA
aHUrqnVNMst8DwQQBkybR1Uu9Ubq9Je61isu3VRXdjrF+Vrq/uZE19zRtIECgYAA
dLX0ofObQsk0KYv3n5jrby3Njm+cfiv/N7y76dSZ2wc+KAMwN0kdDwrPr+3j9NLA
YKERfdBdhKAuNHSTF3S6oxQWJoC9nImxSxp0o5mDTjJso/GKwdu08sdvjOq8dvn
2yok4yw/js1BYDEguivK5PncqSI3MCS8pjdSJWgX+QKBGHiL5fhuysrcMfNdzmCT
eUI1SRlrjTD5Nsd8EzuBIMcxM3OZsko2Tu1UHu8N+1kFqpkkkt4F72c5Xd+5gzWI
o2Fd0U+EZJSWQY0atzvKhDycINai9Zffui01yDdNDJ3SZ9PgXTQdqrwcmjZq1NuC
Kn2jKX3n9ixr9lnizf/tc8Pv
-----END PRIVATE KEY-----"""

# 加载私钥
private_key = serialization.load_pem_private_key(
    private_key_pem.encode('utf-8'),
    password=None,
    backend=default_backend()
)

# 假设这是你的加密数据，它应该是base64编码的

```

```

encrypted_data = encrypted

# 解密数据
'''
decrypted_data = private_key.decrypt(
    encrypted_data,
    padding.OAEP(
        mgf=padding.MGF1(algorithm=hashes.SHA256()),
        algorithm=hashes.SHA256(),
        label=None
    )
)
'''

# 使用 PKCS1v1.5 进行解密
decrypted_data = private_key.decrypt(
    encrypted_data,
    padding.PKCS1v15()
)

print("decrypt plain text:" + decrypted_data.hex())

```

5.1.2 CRYPTO_KEY_FORMAT_BIN_IDENT_PRIVATEKEY_PKCS8

以PKCS8 RSA 2048 private key为例，如下所示，这个PEM私钥证书base64解码后就是CRYPTO_KEY_FORMAT_BIN_IDENT_PRIVATEKEY_PKCS8 数据。

```

-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBAKggwggSkAgEAAoIBAQC7BvECYdRsFJKb
oZKgyfQXBFeUks/Z3fxfstaT3tltwYC59mkIaRlCPx+HoSt0kMkyJbEVLBOswzs
M8cUXClhwdIEb/7En40L+i8PU+ldvVDVuuU1OvdYfZVhdo7IAC7XWJmfSbkGwJbus
Ft1SYzDmh4QQOBAZmd9zvvyLctDhNksfs3694pZDY3az6ktfmvpcSqHe9MyrMHKm
e5AbR6L3apDthiFIH0yqTwmZLZAuvvU8WNna86/h79wwboDOTBKf3N/iOC86s5hh
/rbTxd0EY6D7VukxAbukV2kji1LcgZDjmdnI9wPFewSRUHFp+4LnPAJH1IM9IX4r
LIVXtoknAgMBAAECggEBAIbRSCZCZcxEI833wiIwTSC5n3nTKiaeY5FCY++mL5f26
Y0Dw9HhWL/VX7wrs4P2T1gbL+bJpAfax+1aBNH6+u4/+cYJobQQ7pyb9H7NC4pzy
A7UssEdJpM/16E8H0yZ0Qp2ubaTH4pqkdJZmmh3Dltp20YIh8NBqjv7bnzvNRdC
GJpVvbhPhF+jTnbwrkn98gFN2dOedlQZA5QFs1l8SW3hwUHHg5JpcTTkohYbn91J
sm33HB6U23Defk52CkG008jqb7NwbtfIO020480AoZShC5Hj4paZHfYz4kkql+lh
kdkCr1zR6ZAp7Dziym1ak84g+/590tltjx3bl8TSFgECgYEA9HFNA95zLyJwbOpz
65FnTAAVCOCT4t6Kad76yx5AGvkjuEpuZOKaazFRGIq8H0ha6a5UjIdavknyUNWJ
mC9T6P1TdI6zxgFGGRPKdJHrqPTZN3M9/7bKwwrCMj669eI+o6xiaALDLnspAGgx
UXisyefmKG6/jIQA4M2QmKQl6oECgYEAw96xFlG8IYAv3HM4mfUjiym9CuF+heS2
svMK3SOHvivCQtdvaBFj9SftgyUQMSB4vzRFeCjGBfh60T7kCarrvDi0UAE5mJhi
BCUCFMroxBEWS3kozyX8gh1dkR1c0uzhKbvNvQ5vXPqzxZ+YYJJkyZRd+p/waVZL
j+Awp2m9D6cCgYEA1xiGIYhsqQw5LONCEAc582mbGN2tpJQYsr7Ixed/d3zGdtQI
QmQYZ7Zu1Ji4IWPhJAIQSjYmpKobQ8AzvKpg3YOVHVq5HMJgMY71XPpwma0joyUA
aHUrqnVNMst8DwQQBkybR1Uu9UbQ9Je61isu3VRXdjrf+Vrq/uZE19zRtIECgYAA

```



```
dLX0ofoBqsk0KYv3n5jRby3Njm+cfiv/N7y76dSZ2wc+KAMwN0kdDwrPr+3j9NLa
YKERfdBhdhKAuNHsTF3S6oXQWJoC9nImXSxp0o5mDTjJso/GKwdu08sdvjOq8dvn
2yok4yw/js1BYDEguiVK5PncqsI3MCS8pjdSjWgX+QKBgHiL5fhUysrcMfNdzmCT
eUI1SRlrjTD5Nsd8EzuBIMcxM3OZsko2Tu1UHu8N+1kFqpkKkt4F72c5Xd+5gzWI
o2Fd0U+EZJswQY0atzvKhDycINai9Zffui01yDdNDJ3SZ9PgXTQdqrwcmjZq1NuC
kn2jkx3n9ixr9lnizf/tc8Pv
-----END PRIVATE KEY-----
```

base64解码的脚本如下。

```
import base64

data = '''MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBAgEAAoIBAQC7BvECYdRsFJKb
oZKgYfQFxBfeUks/Z3fxfstAt3tltwYC59mkIaRlCPx+Host0kMkyJbEVLBOSwzs
M8cUXC1hwdIEB/7En40L+i8PU+ldvVDVUu10vDYFzVhdo7IAC7XWJmfsbkGwJbus
Ft1SYZDmh4QQOBAZMd9zvvyLctDhNksfs3694pZDY3az6ktfmvpcSqHe9MyrMHKm
e5AbR6L3apDthiFIH0yqTwmZLZAuvvU8WNa86/h79wwboDOTBKf3N/iOC86s5hh
/rbTxd0EY6D7VukxAbukv2Kji1LcgZDjmdnI9wPFewSRUHF+4LnpAJH1IM9IX4r
LIVXtoknAgMBAAECggEBAIbRSCZCxEI833wiIwTSC5n3nTKiaeY5FCY++mL5f26
Y0Dw9HhWL/vX7wrs4P2T1gbL+bJpAfax+1aBNH6+u4/+cYJobQQ7pyb9H7NC4pzy
A7UssEdJpM/16E8H0yZ0Qp2ubaTH4pqkdJZmmh3D1tp20YIhH8NBqjv7bnzvNRdC
GJpvvbhPhF+jTnbwrkn98gFN2doEdlQZA5QfS1l8Sw3hwUHHg5JpcTTkohYbn91J
sm33HB6U23DefK52CkGO08jQb7NwbtfIO020480AoZShc5Hj4pazHFyZ4kkq1+1h
kdkCr1zR6Zap7Dziym1ak84g+/590tltjx3b18TSFgECgYEA9HFNA95zLyJwb0pz
65FnTAAVCOCT4t6Kad76yx5AGvkjuEpuZOKaazFRGIq8Hoha6a5UjIdavknyUNWJ
mC9T6P1TdI6zxgFGRPKDjHrqpTZN3M9/7bKwrrCMj669eI+o6xiaALDLnspAGgx
UXisYefmKG6/jIQA4M2QmKQ16oECgYEAw96xFlG8IYAv3HM4mfUjiym9CuF+heS2
svMK3SOHvivCQtdvaBfj9SftgyUQMSB4vzRFeCjGBfh60T7kCarrvDi0UAE5mJhi
BCUCFMroxBEWS3kozyX8gh1dkr1couzhKbvNvQ5vXPqzxZ+YYJJkyZRd+p/wavZL
j+Awp2m9D6cCgYEA1xiGIYhsqQw5LONCEAc582mbGN2tpJQYsr7Ixed/d3zGdtQI
QmQYZ7zu1Ji4IWPhJAIQSjymPKobQ8AzvKpg3YOVHVq5HMJgMY71Xppwma0jOyUA
aHUrqnVNMst8DwQQBkybr1Uu9UbQ9Je61isu3VRxdjrF+Vrq/uZE19zRtIECgYAA
dLX0ofoBqsk0KYv3n5jRby3Njm+cfiv/N7y76dSZ2wc+KAMwN0kdDwrPr+3j9NLa
YKERfdBhdhKAuNHsTF3S6oXQWJoC9nImXSxp0o5mDTjJso/GKwdu08sdvjOq8dvn
2yok4yw/js1BYDEguiVK5PncqsI3MCS8pjdSjWgX+QKBgHiL5fhUysrcMfNdzmCT
eUI1SRlrjTD5Nsd8EzuBIMcxM3OZsko2Tu1UHu8N+1kFqpkKkt4F72c5Xd+5gzWI
o2Fd0U+EZJswQY0atzvKhDycINai9Zffui01yDdNDJ3SZ9PgXTQdqrwcmjZq1NuC
kn2jkx3n9ixr9lnizf/tc8Pv'''
data = base64.b64decode(data)
print(data.hex())
print("")
```

得到数据就是 CRYPTO_KE_FORMAT_BIN_IDENT_PRIVATEKEY_PKCS8。

308204be020100300d06092a864886f70d0101010500048204a8308204a40201000282010100bb06f10261d46c7c929ba192a0c9f405c417de524b3f6777d77ecb5a4f7b65b70602e7d9a421a46570fc7e1e84add24324c896c454b04eb16cec33c7145c2d61c1d21e07fec49f8d0bfa2f0f53ed5dbd50d5b94d4e55d605cd585da3b20073b5d62667d26ca81625bbac16d9526330e687841038101931df73befc8b72d0e1364b1fb37ebde296436376b3ea4b5f9afa5c4aa1def4ccab3072a67b901b47a2f76a90ed8621481f4caa4d69992d902ebef53c58d9daf3afe1efd5b06e80ce4c129fdcdfe2382f3ab39861feb6d35ddd0463a0fb55493101b5245769098b52dc8190e399d9c8f703c57964915077cfff82e7a40247d4833d217e2b2c8557b689270203010001028201010086d1482659731108f37df0888c13482e67de74ca89a798e45098f8e98be5fdb6340f0f478562ff557ef0ad2e0fd93d606cbf9b26901f697fb5681347ebbbb8ffe7182686d043ba726fd1fb342e2967203b52cb04749a4cfe5e84f07d32674429dae6da4c7e29aa47496669a1dc396da76d1820787c341aa3bfb6e7cef351742189a55bdb84f845fa34e76d6acae9fdf2014dd9d384765419039405b3597c496de15941c78392697134e4a2161b9fdd49b26df71c1e94db70de7cae760a418e3bc8d06fb3706ed7c838edb4e3cd00a194a10b91e3e296991c5c99e2492a97e96191d282ae5cd1e99029ec3662ca695a2bce20fbfe7dd2d96d8f1ddb97c4d2160102818100f4714d03de732f22566f4a73eb91674c001508e093e2de8a01defac97e401af923b84a6e64e29a6b3151188abc1ce85ae9ae548c875a5649f250d589982f53e8fd53748eb3c601461913ca0e31eba8f4d937733dfffb6ca5b0ac2323ebaf5e23ea3ac626802c32e7b290068315178ac61e7e6286ebf8c8400e0cd9098a425ea8102818100c3deb11651bc21802fdc733899f5238b29bd0ae17e85e4b6b2f30add2387562bc242d76f681163f5216d832510312078bf34457828c605f87ad13ee409aaebbc38b450013998986204251c14cae8c411164b7928cf25fc821d5d2ab95c3aece129bbcbdb0e6f5cfab3c59f98609264c9945dfa9fd669564b8fe030a769bd0fa702818100d718a021886ca90c392ce342100739f3699b18ddada49418b2bec8c5e77f777cc60ed40842641867bcd4d498b82163e12402104a3c8ca4a39b43c033bca3e0dd83951d5ab91cc260318ef55cfa5699ad233b250068752ba96355312b7c0f041004ac9b47552ef546d0f497bad62b2edd5457763ac5f95aeafee64497dcd1b4810281801a74b5cea1f39b42c934298bf79f98d16f2dcd8e6f9c7e2bfff37bcbbe9d499db073e28033037491d0f0acfafe3f4d2da60a1117dd061761280b8d1ec4c5dd2ea85d0589a02f67226c52c69d28e660d38c9b0efc62b07543bcb1dbe33aaf1dbe7db2a0ae32c3f8ecd41603120ba254ae4f35caac2373024bca6376c256817f9028180788be5f854cacadc31f35dce609379423549196b8d30f936c77c133b8120c7313373994a4a364eed541eef0dfb5905aa990a2ade05ef67395ddfb9833c08a3615dd14f84649b16418d1ab73bca843c9c20d022f597df522d35c8374d0c9dd267d3e05d341daab59c9a366ad4db822a7da3917de7f625ebf65362cdfed0bc3ef

上面的数据在芯片内部存储是从低往高地址。对应的代码就是如下数组。

```
const uint8_t aPrivateKeyDat[] = {
    0x30, 0x82, 0x4, 0xbe, 0x2, 0x1, 0x0, 0x30, 0xd, 0x6, 0x9, 0x2a, 0x86, 0x48, 0x86,
    0xf7,
    0xd, 0x1, 0x1, 0x1, 0x5, 0x0, 0x4, 0x82, 0x4, 0xa8, 0x30, 0x82, 0x4, 0xa4, 0x2, 0x1,
    0x0, 0x2, 0x82, 0x1, 0x1, 0x0, 0xbb, 0x6, 0xf1, 0x2, 0x61, 0xd4, 0x6c, 0x7c, 0x92, 0x9b,
    0xa1, 0x92, 0xa0, 0xc9, 0xf4, 0x5, 0xc4, 0x17, 0xde, 0x52, 0x4b, 0x3f, 0x67, 0x77, 0xd7,
    0x7e,
    0xcb, 0x5a, 0x4f, 0x7b, 0x65, 0xb7, 0x6, 0x2, 0xe7, 0xd9, 0xa4, 0x21, 0xa4, 0x65, 0x70,
    0xfc,
    0x7e, 0x1e, 0x84, 0xad, 0xd2, 0x43, 0x24, 0xc8, 0x96, 0xc4, 0x54, 0xb0, 0x4e, 0xb1, 0x6c,
    0xec,
    0x33, 0xc7, 0x14, 0x5c, 0x2d, 0x61, 0xc1, 0xd2, 0x1e, 0x7, 0xfe, 0xc4, 0x9f, 0x8d, 0xb,
    0xfa,
    0x2f, 0xf, 0x53, 0xed, 0x5d, 0xbd, 0x50, 0xd5, 0xb9, 0x4d, 0x4e, 0x55, 0xd6, 0x5, 0xcd,
    0x58,
    0x5d, 0xa3, 0xb2, 0x0, 0x73, 0xb5, 0xd6, 0x26, 0x67, 0xd2, 0x6c, 0xa8, 0x16, 0x25, 0xbb,
    0xac,
    0x16, 0xd9, 0x52, 0x63, 0x30, 0xe6, 0x87, 0x84, 0x10, 0x38, 0x10, 0x19, 0x31, 0xdf, 0x73,
    0xbe,
    0xfc, 0x8b, 0x72, 0xd0, 0xe1, 0x36, 0x4b, 0x1f, 0xb3, 0x7e, 0xbd, 0xe2, 0x96, 0x43, 0x63,
    0x76,
```

0xb3, 0xea, 0x4b, 0x5f, 0x9a, 0xfa, 0x5c, 0x4a, 0xa1, 0xde, 0xf4, 0xcc, 0xab, 0x30, 0x72,
0xa6,
0x7b, 0x90, 0x1b, 0x47, 0xa2, 0xf7, 0x6a, 0x90, 0xed, 0x86, 0x21, 0x48, 0x1f, 0x4c, 0xaa,
0x4d,
0x69, 0x99, 0x2d, 0x90, 0x2e, 0xbe, 0xf5, 0x3c, 0x58, 0xd9, 0xda, 0xf3, 0xaf, 0xe1, 0xef,
0xd5,
0xb0, 0x6e, 0x80, 0xce, 0x4c, 0x12, 0x9f, 0xdc, 0xdf, 0xe2, 0x38, 0x2f, 0x3a, 0xb3, 0x98,
0x61,
0xfe, 0xb6, 0xd3, 0x5d, 0xdd, 0x4, 0x63, 0xa0, 0xfb, 0x55, 0x49, 0x31, 0x1, 0xb5, 0x24,
0x57,
0x69, 0x9, 0x8b, 0x52, 0xdc, 0x81, 0x90, 0xe3, 0x99, 0xd9, 0xc8, 0xf7, 0x3, 0xc5, 0x79,
0x64,
0x91, 0x50, 0x77, 0xcf, 0xfb, 0x82, 0xe7, 0xa4, 0x2, 0x47, 0xd4, 0x83, 0x3d, 0x21, 0x7e,
0x2b,
0x2c, 0x85, 0x57, 0xb6, 0x89, 0x27, 0x2, 0x3, 0x1, 0x0, 0x1, 0x2, 0x82, 0x1, 0x1, 0x0,
0x86, 0xd1, 0x48, 0x26, 0x59, 0x73, 0x11, 0x8, 0xf3, 0x7d, 0xf0, 0x88, 0x8c, 0x13, 0x48,
0x2e,
0x67, 0xde, 0x74, 0xca, 0x89, 0xa7, 0x98, 0xe4, 0x50, 0x98, 0xfb, 0xe9, 0x8b, 0xe5, 0xfd,
0xba,
0x63, 0x40, 0xf0, 0xf4, 0x78, 0x56, 0x2f, 0xf5, 0x57, 0xef, 0xa, 0xd2, 0xe0, 0xfd, 0x93,
0xd6,
0x6, 0xcb, 0xf9, 0xb2, 0x69, 0x1, 0xf6, 0x97, 0xfb, 0x56, 0x81, 0x34, 0x7e, 0xbe, 0xbb,
0x8f,
0xfe, 0x71, 0x82, 0x68, 0x6d, 0x4, 0x3b, 0xa7, 0x26, 0xfd, 0x1f, 0xb3, 0x42, 0xe2, 0x96,
0x72,
0x3, 0xb5, 0x2c, 0xb0, 0x47, 0x49, 0xa4, 0xcf, 0xe5, 0xe8, 0x4f, 0x7, 0xd3, 0x26, 0x74,
0x42,
0x9d, 0xae, 0x6d, 0xa4, 0xc7, 0xe2, 0x9a, 0xa4, 0x74, 0x96, 0x66, 0x9a, 0x1d, 0xc3, 0x96,
0xda,
0x76, 0xd1, 0x82, 0x7, 0x87, 0xc3, 0x41, 0xaa, 0x3b, 0xfb, 0x6e, 0x7c, 0xef, 0x35, 0x17,
0x42,
0x18, 0x9a, 0x55, 0xbd, 0xb8, 0x4f, 0x84, 0x5f, 0xa3, 0x4e, 0x76, 0xd6, 0xac, 0xa9, 0xfd,
0xf2,
0x1, 0x4d, 0xd9, 0xd3, 0x84, 0x76, 0x54, 0x19, 0x3, 0x94, 0x5, 0xb3, 0x59, 0x7c, 0x49,
0x6d,
0xe1, 0x59, 0x41, 0xc7, 0x83, 0x92, 0x69, 0x71, 0x34, 0xe4, 0xa2, 0x16, 0x1b, 0x9f, 0xdd,
0x49,
0xb2, 0x6d, 0xf7, 0x1c, 0x1e, 0x94, 0xdb, 0x70, 0xde, 0x7c, 0xae, 0x76, 0xa, 0x41, 0x8e,
0x3b,
0xc8, 0xd0, 0x6f, 0xb3, 0x70, 0x6e, 0xd7, 0xc8, 0x38, 0xed, 0xb4, 0xe3, 0xcd, 0x0, 0xa1,
0x94,
0xa1, 0xb, 0x91, 0xe3, 0xe2, 0x96, 0x99, 0x1c, 0x5c, 0x99, 0xe2, 0x49, 0x2a, 0x97, 0xe9,
0x61,
0x91, 0xd2, 0x82, 0xae, 0x5c, 0xd1, 0xe9, 0x90, 0x29, 0xec, 0x36, 0x62, 0xca, 0x69, 0x5a,
0x2b,
0xce, 0x20, 0xfb, 0xfe, 0x7d, 0xd2, 0xd9, 0x6d, 0x8f, 0x1d, 0xdb, 0x97, 0xc4, 0xd2, 0x16,
0x1,
0x2, 0x81, 0x81, 0x0, 0xf4, 0x71, 0x4d, 0x3, 0xde, 0x73, 0x2f, 0x22, 0x56, 0x6f, 0x4a,
0x73,
0xeb, 0x91, 0x67, 0x4c, 0x0, 0x15, 0x8, 0xe0, 0x93, 0xe2, 0xde, 0x8a, 0x1, 0xde, 0xfa,
0xc9,
0x7e, 0x40, 0x1a, 0xf9, 0x23, 0xb8, 0x4a, 0x6e, 0x64, 0xe2, 0x9a, 0x6b, 0x31, 0x51, 0x18,
0x8a,

0xbc, 0x1c, 0xe8, 0x5a, 0xe9, 0xae, 0x54, 0x8c, 0x87, 0x5a, 0x56, 0x49, 0xf2, 0x50, 0xd5,
0x89,
0x98, 0x2f, 0x53, 0xe8, 0xfd, 0x53, 0x74, 0x8e, 0xb3, 0xc6, 0x1, 0x46, 0x19, 0x13, 0xca,
0xe,
0x31, 0xeb, 0xa8, 0xf4, 0xd9, 0x37, 0x73, 0x3d, 0xff, 0xb6, 0xca, 0x5b, 0xa, 0xc2, 0x32,
0x3e,
0xba, 0xf5, 0xe2, 0x3e, 0xa3, 0xac, 0x62, 0x68, 0x2, 0xc3, 0x2e, 0x7b, 0x29, 0x0, 0x68,
0x31,
0x51, 0x78, 0xac, 0x61, 0xe7, 0xe6, 0x28, 0x6e, 0xbf, 0x8c, 0x84, 0x0, 0xe0, 0xcd, 0x90,
0x98,
0xa4, 0x25, 0xea, 0x81, 0x2, 0x81, 0x81, 0x0, 0xc3, 0xde, 0xb1, 0x16, 0x51, 0xbc, 0x21,
0x80,
0x2f, 0xdc, 0x73, 0x38, 0x99, 0xf5, 0x23, 0x8b, 0x29, 0xbd, 0xa, 0xe1, 0x7e, 0x85, 0xe4,
0xb6,
0xb2, 0xf3, 0xa, 0xdd, 0x23, 0x87, 0x56, 0x2b, 0xc2, 0x42, 0xd7, 0x6f, 0x68, 0x11, 0x63,
0xf5,
0x21, 0x6d, 0x83, 0x25, 0x10, 0x31, 0x20, 0x78, 0xbf, 0x34, 0x45, 0x78, 0x28, 0xc6, 0x5,
0xf8,
0x7a, 0xd1, 0x3e, 0xe4, 0x9, 0xaa, 0xeb, 0xbc, 0x38, 0xb4, 0x50, 0x1, 0x39, 0x98, 0x98,
0x62,
0x4, 0x25, 0x1c, 0x14, 0xca, 0xe8, 0xc4, 0x11, 0x16, 0x4b, 0x79, 0x28, 0xcf, 0x25, 0xfc,
0x82,
0x1d, 0x5d, 0x2a, 0xb9, 0x5c, 0x3a, 0xec, 0xe1, 0x29, 0xbb, 0xcd, 0xbd, 0xe, 0x6f, 0x5c,
0xfa,
0xb3, 0xc5, 0x9f, 0x98, 0x60, 0x92, 0x64, 0xc9, 0x94, 0x5d, 0xfa, 0x9f, 0xd6, 0x69, 0x56,
0x4b,
0x8f, 0xe0, 0x30, 0xa7, 0x69, 0xbd, 0xf, 0xa7, 0x2, 0x81, 0x81, 0x0, 0xd7, 0x18, 0xa0,
0x21,
0x88, 0x6c, 0xa9, 0xc, 0x39, 0x2c, 0xe3, 0x42, 0x10, 0x7, 0x39, 0xf3, 0x69, 0x9b, 0x18,
0xdd,
0xad, 0xa4, 0x94, 0x18, 0xb2, 0xbe, 0xc8, 0xc5, 0xe7, 0x7f, 0x77, 0x7c, 0xc6, 0xe, 0xd4,
0x8,
0x42, 0x64, 0x18, 0x67, 0xbc, 0xd4, 0xd4, 0x98, 0xb8, 0x21, 0x63, 0xe1, 0x24, 0x2, 0x10,
0x4a,
0x3c, 0x8c, 0xa4, 0xa3, 0x9b, 0x43, 0xc0, 0x33, 0xbc, 0xa3, 0xe0, 0xdd, 0x83, 0x95, 0x1d,
0x5a,
0xb9, 0x1c, 0xc2, 0x60, 0x31, 0x8e, 0xf5, 0x5c, 0xfa, 0x56, 0x99, 0xad, 0x23, 0x3b, 0x25,
0x0,
0x68, 0x75, 0x2b, 0xa9, 0x63, 0x55, 0x31, 0x2b, 0x7c, 0xf, 0x4, 0x10, 0x4, 0xac, 0x9b,
0x47,
0x55, 0x2e, 0xf5, 0x46, 0xd0, 0xf4, 0x97, 0xba, 0xd6, 0x2b, 0x2e, 0xdd, 0x54, 0x57, 0x76,
0x3a,
0xc5, 0xf9, 0x5a, 0xea, 0xfe, 0xe6, 0x44, 0x97, 0xdc, 0xd1, 0xb4, 0x81, 0x2, 0x81, 0x80,
0x1a,
0x74, 0xb5, 0xce, 0xa1, 0xf3, 0x9b, 0x42, 0xc9, 0x34, 0x29, 0x8b, 0xf7, 0x9f, 0x98, 0xd1,
0x6f,
0x2d, 0xcd, 0x8e, 0x6f, 0x9c, 0x7e, 0x2b, 0xff, 0x37, 0xbc, 0xbb, 0xe9, 0xd4, 0x99, 0xdb,
0x7,
0x3e, 0x28, 0x3, 0x30, 0x37, 0x49, 0x1d, 0xf, 0xa, 0xcf, 0xaf, 0xed, 0xe3, 0xf4, 0xd2,
0xda,
0x60, 0xa1, 0x11, 0x7d, 0xd0, 0x61, 0x76, 0x12, 0x80, 0xb8, 0xd1, 0xec, 0x4c, 0x5d, 0xd2,
0xea,

```

0x85, 0xd0, 0x58, 0x9a, 0x2, 0xf6, 0x72, 0x26, 0xc5, 0x2c, 0x69, 0xd2, 0x8e, 0x66, 0xd,
0x38,
0xc9, 0xb0, 0xef, 0xc6, 0x2b, 0x7, 0x54, 0x3b, 0xcb, 0x1d, 0xbe, 0x33, 0xaa, 0xf1, 0xdb,
0xe7,
0xdb, 0x2a, 0xa, 0xe3, 0x2c, 0x3f, 0x8e, 0xcd, 0x41, 0x60, 0x31, 0x20, 0xba, 0x25, 0x4a,
0xe4,
0xf3, 0x5c, 0xaa, 0xc2, 0x37, 0x30, 0x24, 0xbc, 0xa6, 0x37, 0x6c, 0x25, 0x68, 0x17, 0xf9,
0x2,
0x81, 0x80, 0x78, 0x8b, 0xe5, 0xf8, 0x54, 0xca, 0xca, 0xdc, 0x31, 0xf3, 0x5d, 0xce, 0x60,
0x93,
0x79, 0x42, 0x35, 0x49, 0x19, 0x6b, 0x8d, 0x30, 0xf9, 0x36, 0xc7, 0x7c, 0x13, 0x3b, 0x81,
0x20,
0xc7, 0x31, 0x33, 0x73, 0x99, 0x4a, 0x4a, 0x36, 0x4e, 0xed, 0x54, 0x1e, 0xef, 0xd, 0xfb,
0x59,
0x5, 0xaa, 0x99, 0xa, 0x2a, 0xde, 0x5, 0xef, 0x67, 0x39, 0x5d, 0xdf, 0xb9, 0x83, 0x3c,
0x8,
0xa3, 0x61, 0x5d, 0xd1, 0x4f, 0x84, 0x64, 0x9b, 0x16, 0x41, 0x8d, 0x1a, 0xb7, 0x3b, 0xca,
0x84,
0x3c, 0x9c, 0x20, 0xd0, 0x22, 0xf5, 0x97, 0xdf, 0x52, 0x2d, 0x35, 0xc8, 0x37, 0x4d, 0xc,
0x9d,
0xd2, 0x67, 0xd3, 0xe0, 0x5d, 0x34, 0x1d, 0xaa, 0xb5, 0x9c, 0x9a, 0x36, 0x6a, 0xd4, 0xdb,
0x82,
0x2a, 0x7d, 0xa3, 0x91, 0x7d, 0xe7, 0xf6, 0x25, 0xeb, 0xf6, 0x53, 0x62, 0xcd, 0xff, 0xed,
0xb,
0xc3, 0xef
};

```

转换的脚本如下。

```

def split():
    data =
bytes.fromhex("308204be020100300d06092a864886f70d0101010500048204a8308204a402010002820101
00bb06f10261d46c7c929ba192a0c9f405c417de524b3f6777d77ecb5a4f7b65b70602e7d9a421a46570fc7e1
e84add24324c896c454b04eb16cec33c7145c2d61c1d21e07fec49f8d0bfa2f0f53ed5dbd50d5b94d4e55d605
cd585da3b20073b5d62667d26ca81625bbac16d9526330e687841038101931df73befc8b72d0e1364b1fb37eb
de296436376b3ea4b5f9afa5c4aa1def4ccab3072a67b901b47a2f76a90ed8621481f4caa4d69992d902ebef5
3c58d9daf3afe1efd5b06e80ce4c129fdcdf2382f3ab39861feb6d35ddd0463a0fb55493101b5245769098b5
2dc8190e399d9c8f703c57964915077cfff82e7a40247d4833d217e2b2c8557b6892702030100010282010100
86d1482659731108f37df0888c13482e67de74ca89a798e45098f98be5fdb6340f0f478562ff557ef0ad2e
0fd93d606cbf9b26901f697fb5681347ebbb8ffe7182686d043ba726fd1fb342e2967203b52cb04749a4cfe5
e84f07d32674429dae6da4c7e29aa47496669a1dc396da76d1820787c341aa3bfb6e7cef351742189a55bdb84
f845fa34e76d6aca9fdf2014dd9d384765419039405b3597c496de15941c78392697134e4a2161b9fdd49b26d
f71c1e94db70de7cae760a418e3bc8d06fb3706ed7c838edb4e3cd00a194a10b91e3e296991c5c99e2492a97e
96191d282ae5cd1e99029ec3662ca695a2bce20fbfe7dd2d96d8f1ddb97c4d2160102818100f4714d03de732f
22566f4a73eb91674c001508e093e2de8a01defac97e401af923b84a6e64e29a6b3151188abc1ce85ae9ae548
c875a5649f250d589982f53e8fd53748eb3c601461913ca0e31eba8f4d937733dfffb6ca5b0ac2323ebaf5e23e
a3ac626802c32e7b290068315178ac61e7e6286ebf8c8400e0cd9098a425ea8102818100c3deb11651bc21802
fdc733899f5238b29bd0ae17e85e4b6b2f30add2387562bc242d76f681163f5216d832510312078bf34457828
c605f87ad13ee409aaebbc38b450013998986204251c14cae8c411164b7928cf25fc821d5d2ab95c3aece129b
bcdbd0e6f5cfab3c59f98609264c9945dfa9fd669564b8fe030a769bd0fa702818100d718a021886ca90c392c
e342100739f3699b18ddada49418b2bec8c5e77f777cc60ed40842641867bcd4d498b82163e12402104a3c8ca
4a39b43c033bca3e0dd83951d5ab91cc260318ef55cfa5699ad233b250068752ba96355312b7c0f041004ac9b
47552ef546d0f497bad62b2edd5457763ac5f95aeafee64497dcd1b4810281801a74b5cea1f39b42c934298bf
79f98d16f2dcd8e6f9c7e2bff37bcbbe9d499db073e28033037491d0f0acfafede3f4d2da60a1117dd0617612
80b8d1ec4c5dd2ea85d0589a02f67226c52c69d28e660d38c9b0efc62b07543bcb1dbe33aaf1dbe7db2a0ae32
c3f8ecd41603120ba254ae4f35caac2373024bca6376c256817f9028180788be5f854cacadc31f35dce609379
423549196b8d30f936c77c133b8120c7313373994a4a364eed541eef0dfb5905aa990a2ade05ef67395ddfb98
33c08a3615dd14f84649b16418d1ab73bca843c9c20d022f597df522d35c8374d0c9dd267d3e05d341daab59c
9a366ad4db822a7da3917de7f625ebf65362cdf fed0bc3ef")
    #print(Swapu32ByteOrder(data).hex())
    index = 0
    for onedata in data:
        print("0x%x, " % onedata, end="")
        index = index + 1
        if (index % 16) == 0:
            print("")

```

5.1.2.1 配置

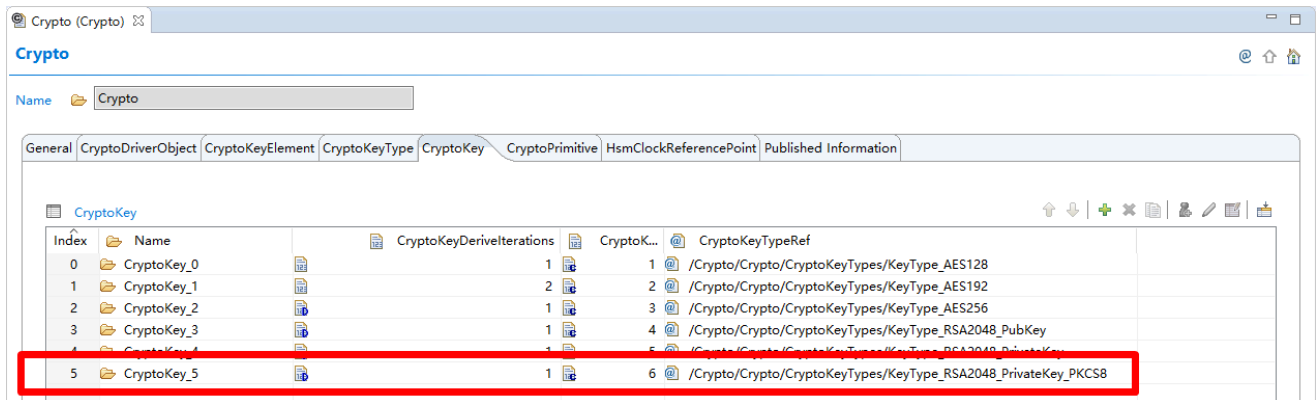
因为证书长度不固定，所以buffer按照最大长度设置，RSA2048密钥一般size是 $(2048/8)*5.5=1408$ 字节，为了保险起见，配置为2048字节。填充的数据不会填满buffer，所以选中CryptoKeyElementAllowPartialAccess。

Crypto (Crypto)									
Crypto									
Name Crypto									
General CryptoDriverObject CryptoKeyElement CryptoKeyType CryptoKey CryptoPrimitive HsmClockReferencePoint Published Information									
CryptoKeyElement									
Index	Name	CryptoKeyEl...	CryptoKeyElementFormat	CryptoKeyE...	CryptoK...	Crypto...	Crypto...	CryptoKeyElementR	
0	CryptoKeyElement_0_AES128_KEY		CRYPTO_KE_FORMAT_BIN_OCTET	1	1			CRYPTO_RA_ALLOWE	
1	CryptoKeyElement_1_AES128_IV		CRYPTO_KE_FORMAT_BIN_OCTET	5	1			CRYPTO_RA_ALLOWE	
2	CryptoKeyElement_2_AES192_KEY		CRYPTO_KE_FORMAT_BIN_OCTET	1	1			CRYPTO_RA_ALLOWE	
3	CryptoKeyElement_3_AES256_KEY		CRYPTO_KE_FORMAT_BIN_OCTET	1	1			CRYPTO_RA_ALLOWE	
4	CryptoKeyElement_4_RSA2048_N_KEY		CRYPTO_KE_FORMAT_BIN_OCTET	1001	1			CRYPTO_RA_ALLOWE	
5	CryptoKeyElement_5_RSA2048_E_KEY		CRYPTO_KE_FORMAT_BIN_OCTET	1000	1			CRYPTO_RA_ALLOWE	
6	CryptoKeyElement_6_RSA2048_D_KEY		CRYPTO_KE_FORMAT_BIN_OCTET	1000	1			CRYPTO_RA_ALLOWE	
7	CryptoKeyElement_7_RSA2048_PRIVATE_PKCS8_KEY		CRYPTO_KE_FORMAT_BIN_IDENT...	1	1			CRYPTO_RA_ALLOWE	

Crypto (Crypto)	
CryptoKeyElement	
Name CryptoKeyElement_7_RSA2048_PRIVATE_PKCS8_KEY	
General	
CryptoKeyElementAllowPartialAccess	☒
CryptoKeyElementFormat	E_FORMAT_BIN_IDENT_PRIVATEKEY_PKCS8
CryptoKeyElementId (1 -> 4294967295)	1
CryptoKeyElementInitValue	1
CryptoKeyElementPersist	☐
CryptoKeyElementReadAccess	CRYPTO_RA_ALLOWED
CryptoKeyElementSize (1 -> 4294967295)	2048
CryptoKeyElementWriteAccess	CRYPTO_WA_ALLOWED
CryptoKeyElementVirtualTargetRef	
CryptoKeyStoreType	RAM_KEY
CryptoKeyNvmKeyId (1 -> 4294967295)	8

Crypto (Crypto)									
Crypto									
Name Crypto									
General CryptoDriverObject CryptoKeyElement CryptoKeyType CryptoKey CryptoPrimitive HsmClockReferencePoint Published Information									
CryptoKeyType									
Index	Name								
0	KeyType_AES128								
1	KeyType_AES192								
2	KeyType_AES256								
3	KeyType_RSA2048_PubKey								
4	KeyType_RSA2048_PrivateKey								
5	KeyType_RSA2048_PrivateKey_PKCS8								

Crypto (Crypto)	
CryptoKeyType	
Name KeyType_RSA2048_PrivateKey_PKCS8	
CryptoKeyElementRef	
CryptoKeyElementRef	
Index	CryptoKeyElementRef
0	/Crypto/Crypto/CryptoKeyElements/CryptoKeyElement_7_RSA2048_PRIVATE_PKCS8_KEY



5.1.2.2 使用

```
/*
PKCS1 RSA 2048 private key
-----BEGIN RSA PRIVATE KEY-----
MIIEPaIBAACAQEAuwbxAmHUBHySm6GSomn0BcQX3lJLP2d3137Lwk97ZbcGAufZ
pCGkZXD8fh6ErdJDJMiWxFswTrFs7DPHFFwtYCHSHgf+xJ+NC/ovD1PtXb1Q1b1N
TlXWBC1YXaOyAH011iZn0myoFiw7rBbZUMw5oeEEDgQGTHfc778i3LQ4TZLH7N+
veKWQ2N2s+pLX5r6XEqh3vTMqzBypnuQG0ei92qQ7YYhsB9Mqk1pmS2QLr71PFjZ
2vov4e/VsG6AzkwSn9zf4jgvOrOYYf62013dBGog+1VJMQG1JFdpCYtS3IGQ45nZ
yPCdXlkkVB3z/uc56QCR9SDPSF+KyyFV7aJjWIDAQABAOIBAQCgOUgmWXMRCpN9
8Iime0guZ950yomnmORQmPvpi+x9umNA8PR4Vi/1v+8K0ud9k9Ygy/myaQH2l/tw
gTR+vruP/ngCaG0E06cm/R+zQuKWcg01LLBHSaTP5ehPB9MmdEKdrn2kx+KapHSW
Zpodw5badtGCB4fDQao7+2587zUXQhiavb24T4Rfo0521qyp/fIBTdnThHZUGQOU
BbNZfEl4v1Bx40sAXE05KIWG5/dsbJt9xwe1ntw3nyudgpBjjvIOG+zCG7XyDjt
tOPNAKUoQuR4+KwMRxcmeJJKpfYZHsgq5c0emQKew2YsppwiVoIPv+fdLZbY8d
25fE0hYBAOGBAPRXTQpecy8iVm9Kc+uRZ0wAFQjgk+LeigHe+s1+QBr5I7hkBmTi
mmsxURiKvBzowumuVIyHw1ZJ8lDViZgvU+j9U3S0s8YBRhkTyg4x66j02TdZPf+2
y1skwji+uvXiPqOsYmgCwy57KQBoMVf4rGhN5ihuv4yEAODNkjiKJeqBAoGBAMPe
sRZRvCGAL9xz0Jn1I4spvQrhfoXktrLzCt0jh1YrwlXb2gRY/UhbyMlEDEgeL80
RXgogxG4etE+5Amq67w4tFABOziYYgQlHBTk6MQRfkt5KM8l/IIdXSq5XDrs4Sm7
zb00blz6s8wfmGCSZMmUxfqf1m1ws4/gMKdpvQ+nAoGBANcyoCGIBkkmOSzjQhAH
OfNpmxjdrasUGLK+yMxnf3d8xg7UCEJkGGe81NSYUCFj4SQCEE08jksjm0PAM7yj
4N2DlR1aurZCYDGO9Vz6VpmtIzslAGh1K6ljVTERfA8EEASm0dVLvVG0PSXutYr
Lt1UV3Y6xf1a6v7mRjfc0bSBAoGAGNs1zqHzm0LJNCmL95+Y0W8tzY5vnH4r/ze8
u+nUmdSHpiGDMddJHQ8Kz6/t4/TS2mChEX3QYXYSgLjr7Exd0uqF0FiaAvZyJSUS
adKOZg04ybDvxiSHVDvLHb4zqvHb59sqCuMsP47NQWAXILOlSuTzXKrcNzAkVky3
bcVoF/kCgYB4i+x4VMrK3DHZxc5gk3lCNUkZa40w+TbHfBM7gSDHMTNzmUpKNk7t
VB7vDftZBaqZCireBe9NOV3fuYM8CKNhxdFPhGsBfKGNgrc7yoQ8nCDQIvWX31It
Ncg3TQyd0mft4F00Haq1nJo2atTbgiP9o5F95/Y16/ZTys3/7QvD7w==
-----END RSA PRIVATE KEY-----
```

```
PKCS8 RSA 2048 private key
-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBKgwggSkAgEAAoIBAQC7BVECYdRsFJKb
oZKgyfQFxBfeUks/Z3fxfstaT3tltwYC59mkIarlCPx+Host0kMkyJbEVLBOSWzs
M8cUXC1hwdIEB/7En40L+i8PU+IdvVDVuu10vdYFzVhdo7Iac7XWJmfsbkgWJbus
FtlSYZdmh4QQOBAZMd9zvvyLctDhNksfs3694pZDY3az6ktfmvpcSqHe9MyrMHK
```



```
e5AbR6L3apDthiFIH0yqTwmZLZAuvvU8WNna86/h79wwboDOTBKf3N/iOC86s5hh
/rbTxd0EY6D7VukxAbukv2Kji1LcgZDjmdnI9wPFewSRUHfP+4LnpAJH1IM9IX4r
LIVXtoknAgMBAECggEBAIbRSCZCxcEI833wiIwTSC5n3nTKiaeY5FCY++mL5f26
Y0Dw9HhWL/VX7wrs4P2T1gbL+bJpAfax+1aBNH6+u4/+cYJobQQ7pyb9H7NC4pzy
A7UssEdJpM/16E8H0yZ0Qp2ubaTH4pqkdJZmmh3D1tp20YIHh8NBqjv7bnzvNRdC
GJpvvbhPhF+jTnbwrKn98gFN2d0Ed1QZA5QFs118Sw3hwUHHg5JpcTTkohYbn91J
sm33HB6U23DefK52CkG008jQb7NwbtfIO020480AoZShC5Hj4pazHFyZ4kkq1+1h
kdkCr1zR6ZAp7DZiym1ak84g+/590t1tjx3b18TSFgECgYEA9HFNA95zLyJwb0pz
65FnTAAVCOCT4t6Kad76yx5AGvkjuEpuZOKaazFRGIq8H0ha6a5UjIdavknyUNWJ
mC9T6P1TdI6zxgFGRPKDjHrqpTZN3M9/7bkWwrcMj669eI+o6xiaALDLnspAGgx
UXisYefmKG6/jIQA4M2QmKQ16oECgYEAw96xFlG8IYAv3HM4mfUjiym9CuF+heS2
svMK3SOHVivCQtdvaBFj9SftgyUQMSB4vzRFeCjGBfh60T7kCarrvDi0UAE5mJhi
BCUCFMroxBEWS3kozyX8gh1dkr1couzhKbvNvQ5vXPqzxZ+YYJJkyZrd+p/wavZL
j+Awp2m9D6cCgYEA1xiGIYhsqQw5LONCEAc582mbGN2tpJQYsr7Ixed/d3zGdtQI
QmQYZ7zU1Ji4IWPhJAIQSjYmpKobQ8AzvKpg3YOVHVq5HJMjgMY71XPpwma0joyUA
aHUrqnVNMst8DwQQBkybr1Uu9UbQ9Je61isu3VRxdjrf+Vrq/uzE19zRtIECgYAA
dLXOofObqsk0KYv3n5jrby3Njm+cfiv/N7y76dSZ2wc+KAMwN0kdDwrPr+3j9NLa
YKERfdBhdhKAuNHsTF3S6oxQWJoC9nImxSxp0o5MDTjJso/GkwdU08sdvjOq8dvn
2yok4yw/js1BYDEguivK5PncqsI3MCS8pjdSJwgX+QKBgHiL5fhuysrcmFNdzmCT
eUI1SR1rjTD5nsd8EzUBIMcxM3OZsko2Tu1UHu8N+1kFqpkKKt4F72c5Xd+5gzWI
o2Fd0U+EZJSWQY0atzvKhDycINAI9Zffui01yDdNDJ3SZ9PgXTQdqrwcmjZq1NuC
Kn2jKX3n9ixr91Nizf/tc8Pv
-----END PRIVATE KEY-----
```

PKCS8 RSA 2048 public key

-----BEGIN PUBLIC KEY-----

```
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAuwbxAmHUBHySm6GSomN0
BcQX3lJLPd3137Lwk97ZbcGAufZpCGkZXD8fh6ErdJDJMiwxFSwTrFs7DPHFFwt
YCHSHgf+xJ+NC/ovD1PtXb1Q1b1NT1XWBC1YXaOyAH011iZn0myoFiw7rBbZUmMw
5oeEEDgQGTHfc778i3LQ4TZLH7N+veKWQ2N2s+PLX5r6XEqh3vTMqzBypnuQG0ei
92qQ7YYhSB9Mqk1pms2QLr71PFjZ2vov4e/VsG6AzkwSn9zf4jgvorOYYf62013d
BG0g+1VJMQG1JFdpCYts3IGQ45nZyPCDX1kkVB3z/uc56QCR9SDPSF+KyyFV7aJ
JwIDAQAB
```

-----END PUBLIC KEY-----

after base64 decode is

PKCS1 RSA 2048 private key

```
308204a40201000282010100bb06f102
61d46c7c929ba192a0c9f405c417de52
4b3f6777d77ecb5a4f7b65b70602e7d9
a421a46570fc7e1e84add24324c896c4
54b04eb16cec33c7145c2d61c1d21e07
fec49f8d0bfa2f0f53ed5dbd50d5b94d
4e55d605cd585da3b20073b5d62667d2
6ca81625bbac16d9526330e687841038
101931df73befc8b72d0e1364b1fb37e
bde296436376b3ea4b5f9afa5c4aa1de
f4ccab3072a67b901b47a2f76a90ed86
21481f4caa4d69992d902ebef53c58d9
daf3afe1efd5b06e80ce4c129fdcdfe2
382f3ab39861feb6d35ddd0463a0fb55
493101b5245769098b52dc8190e399d9
```

c8f703c57964915077cffb82e7a40247
d4833d217e2b2c8557b6892702030100
01028201010086d1482659731108f37d
f0888c13482e67de74ca89a798e45098
fbe98be5fdbab6340f0f478562ff557ef
0ad2e0fd93d606cbf9b26901f697fb56
81347ebebb8ffe7182686d043ba726fd
1fb342e2967203b52cb04749a4cfe5e8
4f07d32674429dae6da4c7e29aa47496
669a1dc396da76d1820787c341aa3bfb
6e7cef351742189a55bdb84f845fa34e
76d6aca9fdf2014dd9d3847654190394
05b3597c496de15941c78392697134e4
a2161b9fdd49b26df71c1e94db70de7c
ae760a418e3bc8d06fb3706ed7c838ed
b4e3cd00a194a10b91e3e296991c5c99
e2492a97e96191d282ae5cd1e99029ec
3662ca695a2bce20fbfe7dd2d96d8f1d
db97c4d2160102818100f4714d03de73
2f22566f4a73eb91674c001508e093e2
de8a01defac97e401af923b84a6e64e2
9a6b3151188abc1ce85ae9ae548c875a
5649f250d589982f53e8fd53748eb3c6
01461913ca0e31eba8f4d937733dfffb6
ca5b0ac2323ebaf5e23ea3ac626802c3
2e7b290068315178ac61e7e6286ebf8c
8400e0cd9098a425ea8102818100c3de
b11651bc21802fdc733899f5238b29bd
0ae17e85e4b6b2f30add2387562bc242
d76f681163f5216d832510312078bf34
457828c605f87ad13ee409aaebbc38b4
50013998986204251c14cae8c411164b
7928cf25fc821d5d2ab95c3aece129bb
cdbc0e6f5cfab3c59f98609264c9945d
fa9fd669564b8fe030a769bd0fa70281
8100d718a021886ca90c392ce3421007
39f3699b18ddada49418b2bec8c5e77f
777cc60ed40842641867bcd4d498b821
63e12402104a3c8ca4a39b43c033bca3
e0dd83951d5ab91cc260318ef55cfa56
99ad233b250068752ba96355312b7c0f
041004ac9b47552ef546d0f497bad62b
2edd5457763ac5f95aeafee64497dcd1
b4810281801a74b5cea1f39b42c93429
8bf79f98d16f2dcd8e6f9c7e2bff37bc
bbe9d499db073e28033037491d0f0acf
afede3f4d2da60a1117dd061761280b8
d1ec4c5dd2ea85d0589a02f67226c52c
69d28e660d38c9b0efc62b07543bcb1d
be33aaf1dbe7db2a0ae32c3f8ecd4160
3120ba254ae4f35caac2373024bca637
6c256817f9028180788be5f854cacadc
31f35dce609379423549196b8d30f936

c77c133b8120c7313373994a4a364eed
541eef0dfb5905aa990a2ade05ef6739
5ddfb9833c08a3615dd14f84649b1641
8d1ab73bca843c9c20d022f597df522d
35c8374d0c9dd267d3e05d341daab59c
9a366ad4db822a7da3917de7f625ebf6
5362cdf fed0bc3ef

PKCS8 RSA 2048 private key

308204be020100300d06092a864886f7
0d0101010500048204a8308204a40201
000282010100bb06f10261d46c7c929b
a192a0c9f405c417de524b3f6777d77e
cb5a4f7b65b70602e7d9a421a46570fc
7e1e84add24324c896c454b04eb16cec
33c7145c2d61c1d21e07fec49f8d0bfa
2f0f53ed5dbd50d5b94d4e55d605cd58
5da3b20073b5d62667d26ca81625bbac
16d9526330e687841038101931df73be
fc8b72d0e1364b1fb37ebde296436376
b3ea4b5f9afa5c4aa1def4ccab3072a6
7b901b47a2f76a90ed8621481f4caa4d
69992d902ebef53c58d9daf3afe1efd5
b06e80ce4c129fdcdfef2382f3ab39861
feb6d35ddd0463a0fb55493101b52457
69098b52dc8190e399d9c8f703c57964
915077cffb82e7a40247d4833d217e2b
2c8557b6892702030100010282010100
86d1482659731108f37df0888c13482e
67de74ca89a798e45098f9be98be5fdb
6340f0f478562ff557ef0ad2e0fd93d6
06cbf9b26901f697fb5681347ebbb8f
fe7182686d043ba726fd1fb342e29672
03b52cb04749a4cfe5e84f07d3267442
9dae6da4c7e29aa47496669a1dc396da
76d1820787c341aa3bfb6e7cef351742
189a55bdb84f845fa34e76d6aca9fdf2
014dd9d384765419039405b3597c496d
e15941c78392697134e4a2161b9fdd49
b26df71c1e94db70de7cae760a418e3b
c8d06fb3706ed7c838edb4e3cd00a194
a10b91e3e296991c5c99e2492a97e961
91d282ae5cd1e99029ec3662ca695a2b
ce20fbfe7dd2d96d8f1ddb97c4d21601
02818100f4714d03de732f22566f4a73
eb91674c001508e093e2de8a01defac9
7e401af923b84a6e64e29a6b3151188a
bc1ce85ae9ae548c875a5649f250d589
982f53e8fd53748eb3c601461913ca0e
31eba8f4d937733dfffb6ca5b0ac2323e
baf5e23ea3ac626802c32e7b29006831

5178ac61e7e6286ebf8c8400e0cd9098
a425ea8102818100c3deb11651bc2180
2fdc733899f5238b29bd0ae17e85e4b6
b2f30add2387562bc242d76f681163f5
216d832510312078bf34457828c605f8
7ad13ee409aaebbc38b4500139989862
04251c14cae8c411164b7928cf25fc82
1d5d2ab95c3aece129bbcbdbd0e6f5cfa
b3c59f98609264c9945dfa9fd669564b
8fe030a769bd0fa702818100d718a021
886ca90c392ce342100739f3699b18dd
ada49418b2bec8c5e77f777cc60ed408
42641867bcd4d498b82163e12402104a
3c8ca4a39b43c033bca3e0dd83951d5a
b91cc260318ef55cfa5699ad233b2500
68752ba96355312b7c0f041004ac9b47
552ef546d0f497bad62b2edd5457763a
c5f95aeafee64497dcd1b4810281801a
74b5cea1f39b42c934298bf79f98d16f
2dcd8e6f9c7e2bfff37bcbbe9d499db07
3e28033037491d0f0acfafede3f4d2da
60a1117dd061761280b8d1ec4c5dd2ea
85d0589a02f67226c52c69d28e660d38
c9b0efc62b07543bcb1dbe33aaf1dbe7
db2a0ae32c3f8ecd41603120ba254ae4
f35caac2373024bca6376c256817f902
8180788be5f854cacadc31f35dce6093
79423549196b8d30f936c77c133b8120
c7313373994a4a364eed541eef0dfb59
05aa990a2ade05ef67395ddfb9833c08
a3615dd14f84649b16418d1ab73bca84
3c9c20d022f597df522d35c8374d0c9d
d267d3e05d341daab59c9a366ad4db82
2a7da3917de7f625ebf65362cdf fed0b
c3ef

PKCS8 RSA 2048 public key

30820122300d06092a864886f70d0101
0105000382010f003082010a02820101
00bb06f10261d46c7c929ba192a0c9f4
05c417de524b3f6777d77ecb5a4f7b65
b70602e7d9a421a46570fc7e1e84add2
4324c896c454b04eb16cec33c7145c2d
61c1d21e07fec49f8d0bfa2f0f53ed5d
bd50d5b94d4e55d605cd585da3b20073
b5d62667d26ca81625bbac16d9526330
e687841038101931df73befc8b72d0e1
364b1fb37ebde296436376b3ea4b5f9a
fa5c4aa1def4ccab3072a67b901b47a2
f76a90ed8621481f4caa4d69992d902e
bef53c58d9daf3afe1efd5b06e80ce4c
129fdcdfe2382f3ab39861feb6d35ddd
0463a0fb55493101b5245769098b52dc

```
8190e399d9c8f703c57964915077cffb
82e7a40247d4833d217e2b2c8557b689
270203010001
```

```
e=10001
```

```
n=
```

```
bb06f10261d46c7c929ba192a0c9f405
c417de524b3f6777d77ecb5a4f7b65b7
0602e7d9a421a46570fc7e1e84add243
24c896c454b04eb16cec33c7145c2d61
c1d21e07fec49f8d0bfa2f0f53ed5dbd
50d5b94d4e55d605cd585da3b20073b5
d62667d26ca81625bbac16d9526330e6
87841038101931df73befc8b72d0e136
4b1fb37ebde296436376b3ea4b5f9afa
5c4aa1def4ccab3072a67b901b47a2f7
6a90ed8621481f4caa4d69992d902ebe
f53c58d9daf3afe1efd5b06e80ce4c12
9fdcdfe2382f3ab39861feb6d35ddd04
63a0fb55493101b5245769098b52dc81
90e399d9c8f703c57964915077cffb82
e7a40247d4833d217e2b2c8557b68927
```

```
uint8_t aPublicKeyE[]={
```

```
0x01, 0x00, 0x01,
```

```
};
```

```
uint8_t aPublicKeyN[]={
```

```
0xbb, 0x06, 0xf1, 0x02,
```

```
0x61, 0xd4, 0x6c, 0x7c,
```

```
0x92, 0x9b, 0xa1, 0x92,
```

```
0xa0, 0xc9, 0xf4, 0x05,
```

```
0xc4, 0x17, 0xde, 0x52,
```

```
0x4b, 0x3f, 0x67, 0x77,
```

```
0xd7, 0x7e, 0xcb, 0x5a,
```

```
0x4f, 0x7b, 0x65, 0xb7,
```

```
0x06, 0x02, 0xe7, 0xd9,
```

```
0xa4, 0x21, 0xa4, 0x65,
```

```
0x70, 0xfc, 0x7e, 0x1e,
```

```
0x84, 0xad, 0xd2, 0x43,
```

```
0x24, 0xc8, 0x96, 0xc4,
```

```
0x54, 0xb0, 0x4e, 0xb1,
```

```
0x6c, 0xec, 0x33, 0xc7,
```

```
0x14, 0x5c, 0x2d, 0x61,
```

```
0xc1, 0xd2, 0x1e, 0x07,
```

```
0xfe, 0xc4, 0x9f, 0x8d,
```

```
0x0b, 0xfa, 0x2f, 0x0f,
```

```
0x53, 0xed, 0x5d, 0xbd,
```

```
0x50, 0xd5, 0xb9, 0x4d,
```

```
0x4e, 0x55, 0xd6, 0x05,
```

```
0xcd, 0x58, 0x5d, 0xa3,
```

```
0xb2, 0x00, 0x73, 0xb5,
```

```
0xd6, 0x26, 0x67, 0xd2,
```

```
0x6c, 0xa8, 0x16, 0x25,  
0xbb, 0xac, 0x16, 0xd9,  
0x52, 0x63, 0x30, 0xe6,  
0x87, 0x84, 0x10, 0x38,  
0x10, 0x19, 0x31, 0xdf,  
0x73, 0xbe, 0xfc, 0x8b,  
0x72, 0xd0, 0xe1, 0x36,  
0x4b, 0x1f, 0xb3, 0x7e,  
0xbd, 0xe2, 0x96, 0x43,  
0x63, 0x76, 0xb3, 0xea,  
0x4b, 0x5f, 0x9a, 0xfa,  
0x5c, 0x4a, 0xa1, 0xde,  
0xf4, 0xcc, 0xab, 0x30,  
0x72, 0xa6, 0x7b, 0x90,  
0x1b, 0x47, 0xa2, 0xf7,  
0x6a, 0x90, 0xed, 0x86,  
0x21, 0x48, 0x1f, 0x4c,  
0xaa, 0x4d, 0x69, 0x99,  
0x2d, 0x90, 0x2e, 0xbe,  
0xf5, 0x3c, 0x58, 0xd9,  
0xda, 0xf3, 0xaf, 0xe1,  
0xef, 0xd5, 0xb0, 0x6e,  
0x80, 0xce, 0x4c, 0x12,  
0x9f, 0xdc, 0xdf, 0xe2,  
0x38, 0x2f, 0x3a, 0xb3,  
0x98, 0x61, 0xfe, 0xb6,  
0xd3, 0x5d, 0xdd, 0x04,  
0x63, 0xa0, 0xfb, 0x55,  
0x49, 0x31, 0x01, 0xb5,  
0x24, 0x57, 0x69, 0x09,  
0x8b, 0x52, 0xdc, 0x81,  
0x90, 0xe3, 0x99, 0xd9,  
0xc8, 0xf7, 0x03, 0xc5,  
0x79, 0x64, 0x91, 0x50,  
0x77, 0xcf, 0xfb, 0x82,  
0xe7, 0xa4, 0x02, 0x47,  
0xd4, 0x83, 0x3d, 0x21,  
0x7e, 0x2b, 0x2c, 0x85,  
0x57, 0xb6, 0x89, 0x27,  
  
};
```

```
public key encrypt 0x1234567890  
result is  
0b10f5d4b248cb0d9dbfa35a61577a41  
d304f000c182252e3987f00999eebb5a  
cc8a48f7c1d742144a1c998764081954  
aa14621894e0028c40828add31958e4c  
2ea361b6c5374c4c7d0f806701c45e9f  
3b969f3910684c2ce4961ca8d91c48df  
b4f6dd053eef33d7dd7720eabc5407a1  
6e5c2ced26b30d372271d6d27ced4843  
c8fbefd479eea94af367d1e9fe7fd8f9
```

b96a0594afce94250cac47bf005bc8eb
016d4f0675d8ba0393234a12bcc814ef
b9a4d3f3ba125d659557eed598540824
f8b868733fc8bcf79f4bbf22cec3ec7b
d6abc98a2d086fa518abf4f4896006cf
6621bbb9c8afbc8bca88f6cb0c7840b0
1e9f9ca0cfbe63629d5b9326634739c7

*/

```
const uint8_t arSaEsPkcs1v15CipherText[] = {
    0xb, 0x10, 0xf5, 0xd4, 0xb2, 0x48, 0xcb, 0xd, 0x9d, 0xbf, 0xa3, 0x5a, 0x61, 0x57,
    0x7a, 0x41,
    0xd3, 0x4, 0xf0, 0x0, 0xc1, 0x82, 0x25, 0x2e, 0x39, 0x87, 0xf0, 0x9, 0x99, 0xee,
    0xbb, 0x5a,
    0xcc, 0x8a, 0x48, 0xf7, 0xc1, 0xd7, 0x42, 0x14, 0x4a, 0x1c, 0x99, 0x87, 0x64, 0x8,
    0x19, 0x54,
    0xaa, 0x14, 0x62, 0x18, 0x94, 0xe0, 0x2, 0x8c, 0x40, 0x82, 0x8a, 0xdd, 0x31, 0x95,
    0x8e, 0x4c,
    0x2e, 0xa3, 0x61, 0xb6, 0xc5, 0x37, 0x4c, 0x4c, 0x7d, 0xf, 0x80, 0x67, 0x1, 0xc4,
    0x5e, 0x9f,
    0x3b, 0x96, 0x9f, 0x39, 0x10, 0x68, 0x4c, 0x2c, 0xe4, 0x96, 0x1c, 0xa8, 0xd9, 0x1c,
    0x48, 0xdf,
    0xb4, 0xf6, 0xdd, 0x5, 0x3e, 0xef, 0x33, 0xd7, 0xdd, 0x77, 0x20, 0xea, 0xbc, 0x54,
    0x7, 0xa1,
    0x6e, 0x5c, 0x2c, 0xed, 0x26, 0xb3, 0xd, 0x37, 0x22, 0x71, 0xd6, 0xd2, 0x7c, 0xed,
    0x48, 0x43,
    0xc8, 0xfb, 0xef, 0xd4, 0x79, 0xee, 0xa9, 0x4a, 0xf3, 0x67, 0xd1, 0xe9, 0xfe, 0x7f,
    0xd8, 0xf9,
    0xb9, 0x6a, 0x5, 0x94, 0xaf, 0xce, 0x94, 0x25, 0xc, 0xac, 0x47, 0xbf, 0x0, 0x5b,
    0xc8, 0xeb,
    0x1, 0x6d, 0x4f, 0x6, 0x75, 0xd8, 0xba, 0x3, 0x93, 0x23, 0x4a, 0x12, 0xbc, 0xc8,
    0x14, 0xef,
    0xb9, 0xa4, 0xd3, 0xf3, 0xba, 0x12, 0x5d, 0x65, 0x95, 0x57, 0xee, 0xd5, 0x98, 0x54,
    0x8, 0x24,
    0xf8, 0xb8, 0x68, 0x73, 0x3f, 0xc8, 0xbc, 0xf7, 0x9f, 0x4b, 0xbf, 0x22, 0xce, 0xc3,
    0xec, 0x7b,
    0xd6, 0xab, 0xc9, 0x8a, 0x2d, 0x8, 0x6f, 0xa5, 0x18, 0xab, 0xf4, 0xf4, 0x89, 0x60,
    0x6, 0xcf,
    0x66, 0x21, 0xbb, 0xb9, 0xc8, 0xaf, 0xbc, 0x8b, 0xca, 0x88, 0xf6, 0xcb, 0xc, 0x78,
    0x40, 0xb0,
    0x1e, 0x9f, 0x9c, 0xa0, 0xcf, 0xbe, 0x63, 0x62, 0x9d, 0x5b, 0x93, 0x26, 0x63, 0x47,
    0x39, 0xc7
};
```

```
const uint8_t arSaEsPkcs1v15PKCS8PrivateCert[] = {
    0x30, 0x82, 0x4, 0xbe, 0x2, 0x1, 0x0, 0x30, 0xd, 0x6, 0x9, 0x2a, 0x86, 0x48, 0x86,
    0xf7,
    0xd, 0x1, 0x1, 0x1, 0x5, 0x0, 0x4, 0x82, 0x4, 0xa8, 0x30, 0x82, 0x4, 0xa4, 0x2, 0x1,
    0x0, 0x2, 0x82, 0x1, 0x1, 0x0, 0xbb, 0x6, 0xf1, 0x2, 0x61, 0xd4, 0x6c, 0x7c, 0x92, 0x9b,
    0xa1, 0x92, 0xa0, 0xc9, 0xf4, 0x5, 0xc4, 0x17, 0xde, 0x52, 0x4b, 0x3f, 0x67, 0x77, 0xd7,
    0x7e,
```

0xcb, 0x5a, 0x4f, 0x7b, 0x65, 0xb7, 0x6, 0x2, 0xe7, 0xd9, 0xa4, 0x21, 0xa4, 0x65, 0x70,
0xfc,
0x7e, 0x1e, 0x84, 0xad, 0xd2, 0x43, 0x24, 0xc8, 0x96, 0xc4, 0x54, 0xb0, 0x4e, 0xb1, 0x6c,
0xec,
0x33, 0xc7, 0x14, 0x5c, 0x2d, 0x61, 0xc1, 0xd2, 0x1e, 0x7, 0xfe, 0xc4, 0x9f, 0x8d, 0xb,
0xfa,
0x2f, 0xf, 0x53, 0xed, 0x5d, 0xbd, 0x50, 0xd5, 0xb9, 0x4d, 0x4e, 0x55, 0xd6, 0x5, 0xcd,
0x58,
0x5d, 0xa3, 0xb2, 0x0, 0x73, 0xb5, 0xd6, 0x26, 0x67, 0xd2, 0x6c, 0xa8, 0x16, 0x25, 0xbb,
0xac,
0x16, 0xd9, 0x52, 0x63, 0x30, 0xe6, 0x87, 0x84, 0x10, 0x38, 0x10, 0x19, 0x31, 0xdf, 0x73,
0xbe,
0xfc, 0x8b, 0x72, 0xd0, 0xe1, 0x36, 0x4b, 0x1f, 0xb3, 0x7e, 0xbd, 0xe2, 0x96, 0x43, 0x63,
0x76,
0xb3, 0xea, 0x4b, 0x5f, 0x9a, 0xfa, 0x5c, 0x4a, 0xa1, 0xde, 0xf4, 0xcc, 0xab, 0x30, 0x72,
0xa6,
0x7b, 0x90, 0x1b, 0x47, 0xa2, 0xf7, 0x6a, 0x90, 0xed, 0x86, 0x21, 0x48, 0x1f, 0x4c, 0xaa,
0x4d,
0x69, 0x99, 0x2d, 0x90, 0x2e, 0xbe, 0xf5, 0x3c, 0x58, 0xd9, 0xda, 0xf3, 0xaf, 0xe1, 0xef,
0xd5,
0xb0, 0x6e, 0x80, 0xce, 0x4c, 0x12, 0x9f, 0xdc, 0xdf, 0xe2, 0x38, 0x2f, 0x3a, 0xb3, 0x98,
0x61,
0xfe, 0xb6, 0xd3, 0x5d, 0xdd, 0x4, 0x63, 0xa0, 0xfb, 0x55, 0x49, 0x31, 0x1, 0xb5, 0x24,
0x57,
0x69, 0x9, 0x8b, 0x52, 0xdc, 0x81, 0x90, 0xe3, 0x99, 0xd9, 0xc8, 0xf7, 0x3, 0xc5, 0x79,
0x64,
0x91, 0x50, 0x77, 0xcf, 0xfb, 0x82, 0xe7, 0xa4, 0x2, 0x47, 0xd4, 0x83, 0x3d, 0x21, 0x7e,
0x2b,
0x2c, 0x85, 0x57, 0xb6, 0x89, 0x27, 0x2, 0x3, 0x1, 0x0, 0x1, 0x2, 0x82, 0x1, 0x1, 0x0,
0x86, 0xd1, 0x48, 0x26, 0x59, 0x73, 0x11, 0x8, 0xf3, 0x7d, 0xf0, 0x88, 0x8c, 0x13, 0x48,
0x2e,
0x67, 0xde, 0x74, 0xca, 0x89, 0xa7, 0x98, 0xe4, 0x50, 0x98, 0xfb, 0xe9, 0x8b, 0xe5, 0xfd,
0xba,
0x63, 0x40, 0xf0, 0xf4, 0x78, 0x56, 0x2f, 0xf5, 0x57, 0xef, 0xa, 0xd2, 0xe0, 0xfd, 0x93,
0xd6,
0x6, 0xcb, 0xf9, 0xb2, 0x69, 0x1, 0xf6, 0x97, 0xfb, 0x56, 0x81, 0x34, 0x7e, 0xbe, 0xbb,
0x8f,
0xfe, 0x71, 0x82, 0x68, 0x6d, 0x4, 0x3b, 0xa7, 0x26, 0xfd, 0x1f, 0xb3, 0x42, 0xe2, 0x96,
0x72,
0x3, 0xb5, 0x2c, 0xb0, 0x47, 0x49, 0xa4, 0xcf, 0xe5, 0xe8, 0x4f, 0x7, 0xd3, 0x26, 0x74,
0x42,
0x9d, 0xae, 0x6d, 0xa4, 0xc7, 0xe2, 0x9a, 0xa4, 0x74, 0x96, 0x66, 0x9a, 0x1d, 0xc3, 0x96,
0xda,
0x76, 0xd1, 0x82, 0x7, 0x87, 0xc3, 0x41, 0xaa, 0x3b, 0xfb, 0x6e, 0x7c, 0xef, 0x35, 0x17,
0x42,
0x18, 0x9a, 0x55, 0xbd, 0xb8, 0x4f, 0x84, 0x5f, 0xa3, 0x4e, 0x76, 0xd6, 0xac, 0xa9, 0xfd,
0xf2,
0x1, 0x4d, 0xd9, 0xd3, 0x84, 0x76, 0x54, 0x19, 0x3, 0x94, 0x5, 0xb3, 0x59, 0x7c, 0x49,
0x6d,
0xe1, 0x59, 0x41, 0xc7, 0x83, 0x92, 0x69, 0x71, 0x34, 0xe4, 0xa2, 0x16, 0x1b, 0x9f, 0xdd,
0x49,
0xb2, 0x6d, 0xf7, 0x1c, 0x1e, 0x94, 0xdb, 0x70, 0xde, 0x7c, 0xae, 0x76, 0xa, 0x41, 0x8e,
0x3b,

0xc8, 0xd0, 0x6f, 0xb3, 0x70, 0x6e, 0xd7, 0xc8, 0x38, 0xed, 0xb4, 0xe3, 0xcd, 0x0, 0xa1,
0x94,
0xa1, 0xb, 0x91, 0xe3, 0xe2, 0x96, 0x99, 0x1c, 0x5c, 0x99, 0xe2, 0x49, 0x2a, 0x97, 0xe9,
0x61,
0x91, 0xd2, 0x82, 0xae, 0x5c, 0xd1, 0xe9, 0x90, 0x29, 0xec, 0x36, 0x62, 0xca, 0x69, 0x5a,
0x2b,
0xce, 0x20, 0xfb, 0xfe, 0x7d, 0xd2, 0xd9, 0x6d, 0x8f, 0x1d, 0xdb, 0x97, 0xc4, 0xd2, 0x16,
0x1,
0x2, 0x81, 0x81, 0x0, 0xf4, 0x71, 0x4d, 0x3, 0xde, 0x73, 0x2f, 0x22, 0x56, 0x6f, 0x4a,
0x73,
0xeb, 0x91, 0x67, 0x4c, 0x0, 0x15, 0x8, 0xe0, 0x93, 0xe2, 0xde, 0x8a, 0x1, 0xde, 0xfa,
0xc9,
0x7e, 0x40, 0x1a, 0xf9, 0x23, 0xb8, 0x4a, 0x6e, 0x64, 0xe2, 0x9a, 0x6b, 0x31, 0x51, 0x18,
0x8a,
0xbc, 0x1c, 0xe8, 0x5a, 0xe9, 0xae, 0x54, 0x8c, 0x87, 0x5a, 0x56, 0x49, 0xf2, 0x50, 0xd5,
0x89,
0x98, 0x2f, 0x53, 0xe8, 0xfd, 0x53, 0x74, 0x8e, 0xb3, 0xc6, 0x1, 0x46, 0x19, 0x13, 0xca,
0xe,
0x31, 0xeb, 0xa8, 0xf4, 0xd9, 0x37, 0x73, 0x3d, 0xff, 0xb6, 0xca, 0x5b, 0xa, 0xc2, 0x32,
0x3e,
0xba, 0xf5, 0xe2, 0x3e, 0xa3, 0xac, 0x62, 0x68, 0x2, 0xc3, 0x2e, 0x7b, 0x29, 0x0, 0x68,
0x31,
0x51, 0x78, 0xac, 0x61, 0xe7, 0xe6, 0x28, 0x6e, 0xbf, 0x8c, 0x84, 0x0, 0xe0, 0xcd, 0x90,
0x98,
0xa4, 0x25, 0xea, 0x81, 0x2, 0x81, 0x81, 0x0, 0xc3, 0xde, 0xb1, 0x16, 0x51, 0xbc, 0x21,
0x80,
0x2f, 0xdc, 0x73, 0x38, 0x99, 0xf5, 0x23, 0x8b, 0x29, 0xbd, 0xa, 0xe1, 0x7e, 0x85, 0xe4,
0xb6,
0xb2, 0xf3, 0xa, 0xdd, 0x23, 0x87, 0x56, 0x2b, 0xc2, 0x42, 0xd7, 0x6f, 0x68, 0x11, 0x63,
0xf5,
0x21, 0x6d, 0x83, 0x25, 0x10, 0x31, 0x20, 0x78, 0xbf, 0x34, 0x45, 0x78, 0x28, 0xc6, 0x5,
0xf8,
0x7a, 0xd1, 0x3e, 0xe4, 0x9, 0xaa, 0xeb, 0xbc, 0x38, 0xb4, 0x50, 0x1, 0x39, 0x98, 0x98,
0x62,
0x4, 0x25, 0x1c, 0x14, 0xca, 0xe8, 0xc4, 0x11, 0x16, 0x4b, 0x79, 0x28, 0xcf, 0x25, 0xfc,
0x82,
0x1d, 0x5d, 0x2a, 0xb9, 0x5c, 0x3a, 0xec, 0xe1, 0x29, 0xbb, 0xcd, 0xbd, 0xe, 0x6f, 0x5c,
0xfa,
0xb3, 0xc5, 0x9f, 0x98, 0x60, 0x92, 0x64, 0xc9, 0x94, 0x5d, 0xfa, 0x9f, 0xd6, 0x69, 0x56,
0x4b,
0x8f, 0xe0, 0x30, 0xa7, 0x69, 0xbd, 0xf, 0xa7, 0x2, 0x81, 0x81, 0x0, 0xd7, 0x18, 0xa0,
0x21,
0x88, 0x6c, 0xa9, 0xc, 0x39, 0x2c, 0xe3, 0x42, 0x10, 0x7, 0x39, 0xf3, 0x69, 0x9b, 0x18,
0xdd,
0xad, 0xa4, 0x94, 0x18, 0xb2, 0xbe, 0xc8, 0xc5, 0xe7, 0x7f, 0x77, 0x7c, 0xc6, 0xe, 0xd4,
0x8,
0x42, 0x64, 0x18, 0x67, 0xbc, 0xd4, 0xd4, 0x98, 0xb8, 0x21, 0x63, 0xe1, 0x24, 0x2, 0x10,
0x4a,
0x3c, 0x8c, 0xa4, 0xa3, 0x9b, 0x43, 0xc0, 0x33, 0xbc, 0xa3, 0xe0, 0xdd, 0x83, 0x95, 0x1d,
0x5a,
0xb9, 0x1c, 0xc2, 0x60, 0x31, 0x8e, 0xf5, 0x5c, 0xfa, 0x56, 0x99, 0xad, 0x23, 0x3b, 0x25,
0x0,

```

0x68, 0x75, 0x2b, 0xa9, 0x63, 0x55, 0x31, 0x2b, 0x7c, 0xf, 0x4, 0x10, 0x4, 0xac, 0x9b,
0x47,
0x55, 0x2e, 0xf5, 0x46, 0xd0, 0xf4, 0x97, 0xba, 0xd6, 0x2b, 0x2e, 0xdd, 0x54, 0x57, 0x76,
0x3a,
0xc5, 0xf9, 0x5a, 0xea, 0xfe, 0xe6, 0x44, 0x97, 0xdc, 0xd1, 0xb4, 0x81, 0x2, 0x81, 0x80,
0x1a,
0x74, 0xb5, 0xce, 0xa1, 0xf3, 0x9b, 0x42, 0xc9, 0x34, 0x29, 0x8b, 0xf7, 0x9f, 0x98, 0xd1,
0x6f,
0x2d, 0xcd, 0x8e, 0x6f, 0x9c, 0x7e, 0x2b, 0xff, 0x37, 0xbc, 0xbb, 0xe9, 0xd4, 0x99, 0xdb,
0x7,
0x3e, 0x28, 0x3, 0x30, 0x37, 0x49, 0x1d, 0xf, 0xa, 0xcf, 0xaf, 0xed, 0xe3, 0xf4, 0xd2,
0xda,
0x60, 0xa1, 0x11, 0x7d, 0xd0, 0x61, 0x76, 0x12, 0x80, 0xb8, 0xd1, 0xec, 0x4c, 0x5d, 0xd2,
0xea,
0x85, 0xd0, 0x58, 0x9a, 0x2, 0xf6, 0x72, 0x26, 0xc5, 0x2c, 0x69, 0xd2, 0x8e, 0x66, 0xd,
0x38,
0xc9, 0xb0, 0xef, 0xc6, 0x2b, 0x7, 0x54, 0x3b, 0xcb, 0x1d, 0xbe, 0x33, 0xaa, 0xf1, 0xdb,
0xe7,
0xdb, 0x2a, 0xa, 0xe3, 0x2c, 0x3f, 0x8e, 0xcd, 0x41, 0x60, 0x31, 0x20, 0xba, 0x25, 0x4a,
0xe4,
0xf3, 0x5c, 0xaa, 0xc2, 0x37, 0x30, 0x24, 0xbc, 0xa6, 0x37, 0x6c, 0x25, 0x68, 0x17, 0xf9,
0x2,
0x81, 0x80, 0x78, 0x8b, 0xe5, 0xf8, 0x54, 0xca, 0xca, 0xdc, 0x31, 0xf3, 0x5d, 0xce, 0x60,
0x93,
0x79, 0x42, 0x35, 0x49, 0x19, 0x6b, 0x8d, 0x30, 0xf9, 0x36, 0xc7, 0x7c, 0x13, 0x3b, 0x81,
0x20,
0xc7, 0x31, 0x33, 0x73, 0x99, 0x4a, 0x4a, 0x36, 0x4e, 0xed, 0x54, 0x1e, 0xef, 0xd, 0xfb,
0x59,
0x5, 0xaa, 0x99, 0xa, 0x2a, 0xde, 0x5, 0xef, 0x67, 0x39, 0x5d, 0xdf, 0xb9, 0x83, 0x3c,
0x8,
0xa3, 0x61, 0x5d, 0xd1, 0x4f, 0x84, 0x64, 0x9b, 0x16, 0x41, 0x8d, 0x1a, 0xb7, 0x3b, 0xca,
0x84,
0x3c, 0x9c, 0x20, 0xd0, 0x22, 0xf5, 0x97, 0xdf, 0x52, 0x2d, 0x35, 0xc8, 0x37, 0x4d, 0xc,
0x9d,
0xd2, 0x67, 0xd3, 0xe0, 0x5d, 0x34, 0x1d, 0xaa, 0xb5, 0x9c, 0x9a, 0x36, 0x6a, 0xd4, 0xdb,
0x82,
0x2a, 0x7d, 0xa3, 0x91, 0x7d, 0xe7, 0xf6, 0x25, 0xeb, 0xf6, 0x53, 0x62, 0xcd, 0xff, 0xed,
0xb,
0xc3, 0xef
};

```

```

void Bsp_CryptoRsaEspkcs1v15DecryptPKCS8Test(void)
{
    /* Sync */
    #if (CRYPTO_DRIVER_MODE == CRYPTO_SYNC)
        Crypto_JobInfoType tCryptoJobInfo = {
            .jobPriority = 0,
            .jobId       = 1,
        };

        Crypto_PrimitiveInfoType tCryptoPrimitiveInfo = {
            .resultLength = 0,
            .service       = CRYPTO_DECRYPT,
        };
    #endif
}

```

```

        .algorithm.family          = CRYPTO_ALGOFAM_RSA,
        .algorithm.secondaryFamily = CRYPTO_ALGOFAM_NOT_SET,
        .algorithm.keyLength       = 2048,
        .algorithm.mode            = CRYPTO_ALGOMODE_RSAES_PKCS1_v1_5,
};

Crypto_JobPrimitiveInfoType tCryptoJobPrimitiveInfo = {
    .callbackId          = 0,
    .primitiveInfo        = &tCryptoPrimitiveInfo,
    .secureCounterId     = 0,
    .cryIfKeyId          = 0,
    .processingType       = CRYPTO_PROCESSING_SYNC,
    .callbackUpdateNotification = false,
};

u32RsaEspkcs1v15ResultLength = sizeof(aRsaEspkcs1v15PlainText);
Crypto_JobType tCryptoJob = {
    .jobId      = 1,
    .jobState   = CRYPTO_JOBSTATE_IDLE,
    .jobPrimitiveInputOutput.mode = CRYPTO_OPERATIONMODE_SINGLECALL,
    .jobPrimitiveInputOutput.inputPtr      = &aRsaEspkcs1v15CipherText2[0],
    .jobPrimitiveInputOutput.inputLength   = sizeof(aRsaEspkcs1v15CipherText2),
    .jobPrimitiveInputOutput.outputPtr     = &aRsaEspkcs1v15PlainText[0],
    .jobPrimitiveInputOutput.outputLengthPtr = &u32RsaEspkcs1v15ResultLength,
    .jobPrimitiveInfo                      = &tCryptoJobPrimitiveInfo,
    .jobInfo                              = &tCryptoJobInfo,
    .cryptoKeyId                          = CryptoConf_CryptoKey_CryptoKey_5,
};

Crypto_KeyElementSet(CryptoConf_CryptoKey_CryptoKey_5, CRYPTO_KE_CIPHER_KEY, aRsaEspkcs1v15PKCS8PrivateCert, sizeof(aRsaEspkcs1v15PKCS8PrivateCert));
Crypto_KeySetValid(CryptoConf_CryptoKey_CryptoKey_5);
Crypto_ProcessJob(1, &tCryptoJob);

/* Async */
#else
Crypto_JobInfoType tCryptoJobInfo = {
    .jobPriority = 0,
    .jobId      = 1,
};

Crypto_PrimitiveInfoType tCryptoPrimitiveInfo = {
    .resultLength      = 0,
    .service           = CRYPTO_DECRYPT,
    .algorithm.family   = CRYPTO_ALGOFAM_RSA,
    .algorithm.secondaryFamily = CRYPTO_ALGOFAM_NOT_SET,
    .algorithm.keyLength = 2048,
    .algorithm.mode     = CRYPTO_ALGOMODE_RSAES_PKCS1_v1_5,
};

Crypto_JobPrimitiveInfoType tCryptoJobPrimitiveInfo = {
    .callbackId          = 0,

```

```

        .primitiveInfo          = &tCryptoPrimitiveInfo,
        .secureCounterId       = 0,
        .cryIfKeyId            = 0,
        .processingType         = CRYPTO_PROCESSING_ASYNC,
        .callbackUpdateNotification = false,
};

u32RsaEspkcs1v15ResultLength = sizeof(aRsaEspkcs1v15PlainText);
Crypto_JobType tCryptoJob = {
    .jobId      = 1,
    .jobState    = CRYPTO_JOBSTATE_IDLE,
    .jobPrimitiveInputOutput.mode = CRYPTO_OPERATIONMODE_SINGLECALL,
    .jobPrimitiveInputOutput.inputPtr      = &aRsaEspkcs1v15CipherText2[0],
    .jobPrimitiveInputOutput.inputLength   = sizeof(aRsaEspkcs1v15CipherText2),
    .jobPrimitiveInputOutput.outputPtr     = &aRsaEspkcs1v15PlainText[0],
    .jobPrimitiveInputOutput.outputLengthPtr = &u32RsaEspkcs1v15ResultLength,
    .jobPrimitiveInfo                      = &tCryptoJobPrimitiveInfo,
    .jobInfo                              = &tCryptoJobInfo,
    .cryptoKeyId                          = CryptoConf_CryptoKey_CryptoKey_5,
};

Crypto_KeyElementSet(CryptoConf_CryptoKey_CryptoKey_5, CRYPTO_KE_CIPHER_KEY, aRsaEspkcs1v1
5PKCS8PrivateCert, sizeof(aRsaEspkcs1v15PKCS8PrivateCert));
Crypto_KeySetValid(CryptoConf_CryptoKey_CryptoKey_5);
Crypto_ProcessJob(1, &tCryptoJob);
if(E_OK == Crypto_ProcessJob(1, &tCryptoJob))
{
    while (CRYPTO_JOBSTATE_ACTIVE == tCryptoJob.jobState);
}
#endif
}

```

